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Electrical connector structure

Abstract

An electric connector structure includes a base, a tongue piece, a shielding case, a first terminal row, a second terminal row, a middle partition, a first shielding sheet and a second shielding sheet. The shielding case sheathes the base and the tongue. The first terminal row and the second terminal row are disposed on two distal surfaces of the tongue piece. The middle partition is disposed in the tongue piece. The first shielding sheet has a first terminal elastic arm extended toward and abutting against the first terminal row, and a first case elastic arm extended toward and abutting against the shielding case. The second shielding sheet has a second terminal elastic arm extended toward and abutting against the second terminal row, and a second case elastic arm extended toward and abutting against the shielding case. Accordingly, interferences of the electromagnetic wave and the crosstalk are improved.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Technical Field

(1) The present disclosure relates to an electrical connector structure, especially to an electrical connector structure which is easy to be assembled and provided with an advantage of having a high stability during the high-frequency and high-speed transmission.

Description of Related Art

(2) A related-art electrical connector, for example a RJ45 connector, a USB connector or a Display-Port connector, is commonly used in various communication devices to provide a function of transferring an electrical signal and serving as a connection bridge. When the electrical connector is used for retransferring a high-frequency signal, interferences of the electromagnetic wave and the crosstalk are generated, the stability of transferring the current and the signal is affected, thus a middle partition arranged between dual terminal rows of the related-art electrical connector is disposed with a protrusion or a convex block used for abutting against a grounding terminal and a shielding case. As such, the aforesaid problems are avoided.

(3) However, when the conductive terminal and the middle partition are distanced from the shielding case, the middle partition is required to be prolonged to abut against the shielding case, thus the production is complicated and the stability and the strength of the whole structure may easily have problems. Moreover, the related-art middle partition is mounted inside a tongue piece with an injection mounting manner. As such, a new mold has to be provided when the product design is changed or the dimension is slightly adjusted, thus the production procedure is complicated and the production cost is high.

(4) Accordingly, the applicant of the present disclosure has devoted himself for improving the mentioned shortages.

SUMMARY OF THE DISCLOSURE

(5) The present disclosure is to provide an electric connector structure, which has advantages of improving interference of the electromagnetic wave and the crosstalk to increase the stability during the high-frequency and high-speed transmission, and being easy to be assembled to simplify the production procedure and save the production cost.

(6) Accordingly, the present invention provides an electric connector structure, which includes an insulation body, a shielding case, a first terminal row, a second terminal row, a middle partition, a first shielding sheet and a second shielding sheet. The insulation body has a base and a tongue piece. The base has a front end and a rear end. The tongue piece is extended out from the front end. The tongue piece has a first distal surface and a second distal surface oppositely arranged. The shielding case sheathes the insulation body. The first terminal row is inserted in the insulation body from the rear end and disposed on the first distal surface. The first terminal row has a plurality of first grounding terminals. The second terminal row is inserted in the insulation body from the rear end and disposed on the second distal surface. The second terminal row has a plurality of second grounding terminals. The middle partition is accommodated in the tongue piece and located between the first terminal row and the second terminal row. The first shielding sheet is inserted in the base and located between the first terminal row and the shielding case. The first shielding sheet has a plurality of first terminal elastic arms and a first case elastic arm. Each of the first terminal elastic arms is extended from the first shielding sheet toward the first terminal row and abuts against each of the first grounding terminals. The first case elastic arm is extended from the first shielding sheet toward the shielding case and abuts against the shielding case. The second shielding sheet is inserted in the base and located between the second terminal row and the shielding case. The second shielding sheet has a plurality of second terminal elastic arms and a second case elastic arm. Each of the second terminal elastic arms is extended from the second shielding sheet toward the second terminal row and abuts against each of the second grounding terminals. The second case elastic arm is extended from the second shielding sheet toward the shielding case and abuts against the shielding case.

(7) According to one embodiment of the present disclosure, the first shielding sheet has a partition elastic arm, and the partition elastic arm is extended from the first shielding sheet toward the middle partition and abuts against the middle partition.

(8) According to one embodiment of the present disclosure, the base has a first accommodating slot and a first positioning slot, the first shielding sheet has a first positioning sheet, the first shielding sheet is accommodated in the first accommodating slot, and the first positioning sheet is inserted and positioned in the first positioning slot.

(9) According to one embodiment of the present disclosure, there are three first positioning slots and three first positioning sheets, wherein one of the first positioning slots and the other two first positioning slots are disposed at two opposite sides of the first accommodating slot, one of the first positioning sheets and the other two first positioning sheets are disposed at two opposite sides of the first shielding sheet, each of the first positioning sheets is inserted and positioned in each of the first positioning slots.

(10) According to one embodiment of the present disclosure, the base has a second accommodating

slot and a second positioning slot, the second shielding sheet has a second positioning sheet, the second shielding sheet is accommodated in the second accommodating slot, and the second positioning sheet is inserted and positioned in the second positioning slot.

(11) According to one embodiment of the present disclosure, the second accommodating slot and the second positioning slot are in communication and adjacently disposed, the base has a block wall arranged between the second accommodating slot and the second positioning slot, and the block wall blocks a lateral surface of the second positioning sheet.

(12) According to one embodiment of the present disclosure, each of the second terminal elastic arms and the second positioning sheet are disposed at two opposite sides of the second shielding sheet.

(13) According to one embodiment of the present disclosure, the base has a flat slot, the flat slot is inwardly extended from the rear end and disposed in the tongue piece, and the middle partition is inserted in the flat slot.

(14) According to one embodiment of the present disclosure, there are a plurality of first case elastic arms, and each of the first case elastic arms and each of the first terminal elastic arms are alternately arranged.

(15) According to one embodiment of the present disclosure, there are a plurality of second case elastic arms, and each of the second case elastic arms and each of the second terminal elastic arms are alternately arranged.

(16) Advantages achieved by the present disclosure are as follows. According to the electrical connector structure of the disclosure, the interference of the electromagnetic wave and the remote crosstalk is effectively improved through each of the first terminal elastic arms and the first case elastic arm of the first shielding sheet respectively abutting against each of the first grounding terminals and the shielding case, and through each of the second terminal elastic arms and the second case elastic arm of the second shielding sheet respectively abutting against each of the second grounding terminals and the shielding case, thus the stability during the high-frequency and high-speed transmission is increased. Moreover, the first terminal row, the first shielding sheet, the second terminal row and the second shielding sheet are inserted and assembled from the rear end of the base, thus the manufacturing procedure does not require the injection molding and mounting process, the assembling production is simplified and the production cost is reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features of the disclosure believed to be novel are set forth with particularity in the appended claims. The disclosure itself, however, may be best understood by reference to the following detailed description of the disclosure, which describes a number of exemplary embodiments of the disclosure, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective view showing the assembly according to the present disclosure;

(3) FIG. 2 is a perspective exploded view according to the present disclosure;

(4) FIG. 3 is a perspective exploded view showing the insulation body and the first shielding sheet according to the present disclosure;

(5) FIG. 4 is a perspective exploded view showing the insulation body and the second shielding sheet according to the present disclosure;

(6) FIG. 5 is a side cross-sectional view showing the first shielding sheet, the second shielding sheet, the first terminal bus, the second terminal bus and the middle partition according to the present disclosure;

(7) FIG. 6 is a top view showing the shielding case not being disposed according to the present disclosure;

(8) FIG. 7 is a bottom view showing the shielding case not being disposed according to the present disclosure;

(9) FIG. 8 is a cross-sectional view showing the shielding case not being disposed according to the present disclosure; and

(10) FIG. 9 is another cross-sectional view showing the shielding case not being disposed according to the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(11) It is to be understood that the terms for indicating positions and the location relation, for example “front”, “rear”, “left”, “right”, “front end”, “rear end”, “distal end”, “vertical”, “horizontal”, “top end” and “bottom end”, are based on the positions and the location relation disclosed in the drawings, and only used for disclosing the present disclosure and not used for indicating or implying the specified location of the device or the components or the specified structure and operation in certain location, thus the present disclosure is not intended to be limiting.

(12) For example, the terms of “first”, “second”, “third”, “forth” and “fifth” are used for illustrating each unit, component, area, layer and/or part. The component, the unit, the area, the layer and/or the part are not limited by the terms. These terms are only used for separating the element, the assembly, the area, the layer or the part. Unless being clearly indicated according to the whole specification, the terms for example “the first”, “the second”, “the third”, “the fourth” and “the fifth” are not used for implying the order or sequence.

(13) The technical contents of this disclosure will become apparent with the detailed description of embodiments accompanied with the illustration of related drawings as follows. It is intended that the embodiments and drawings disclosed herein are to be considered illustrative rather than restrictive.

(14) The present disclosure provides an electrical connector structure. Please refer to FIG. 1 to FIG. 5, the electrical connector structure includes an insulation body **10**, a shielding case **20**, a first terminal row **30**, a second terminal row **40**, a middle partition **50**, a first shielding sheet **60** and a second shielding sheet **70**.

(15) In some embodiments, the insulation body **10** is formed by a plastic injection molding manner to be in a one-piece form (or integrally formed), here is not intended to be limiting. For example, the insulation body **10** may be made of other insulation materials. The insulation body **10** includes a base **11** and a tongue piece **12**. The base **11** has a front end **111** and a rear end **112** oppositely arranged with a forward/backward manner, and a top part **113** and a bottom part **114** oppositely arranged with an upward/downward manner. The tongue piece **12** is extended from the front end **111** of the base **11**. The tongue piece **12** has a first distal surface **121** and a second distal surface **122** oppositely arranged with an upward/downward manner. In some embodiments, the first distal surface **121** is located above the tongue piece **12**, and the second distal surface **122** is located below the tongue piece **12**. In some other embodiments, the first distal surface **121** may be located below the tongue piece **12**, and the second distal surface **122** may be located above the tongue piece **12**. The base **11** has a flat slot **115**. The flat slot **115** is inwardly extended from the rear end **112** and disposed in the tongue piece **12**.

(16) The shielding case **20** is made of a metal material, for example copper, brass, nickel, silver, steel, tin or an alloy thereof, here is not intended to be limiting. The shielding case **20** sheathes an outer side of the insulation body **10**. In some embodiments, the shielding case **20** covers a top side, a bottom side, a left side, and a right side of the insulation body **10**, thus a front end and a rear end of the insulation body **10** are exposed to make a pair of connectors (not shown in figures) be inserted, and make terminal feet be soldered on a circuit board (not shown in figures). The shielding case **20** has a plurality of soldering feet **21** arranged at a left side and a right side of the bottom part **114** and capable of being soldered on the circuit board.

(17) The first terminal row **30** is inserted in the insulation body **10** from the rear end **112** of the base **11**, and disposed on the first distal surface **121** of the tongue piece **12**. In some embodiments, the

first terminal row **30** may be disposed on the second distal surface **122** of the tongue piece **12**. The first terminal row **30** includes a plurality of first grounding terminals **31**, a plurality of first power terminals and a plurality of first signal terminals. Each of the first grounding terminals **31**, each of the first power terminals and each of the first signal terminals are made of an electrical conducting material.

(18) The second terminal row **40** is inserted in the insulation body **10** from the rear end **112** of the base **11**, and disposed on the second distal surface **122** of the tongue piece **12**. In some embodiments, the second terminal row **40** may be disposed on the first distal surface **122** of the tongue piece **12**. The second terminal row **40** includes a plurality of second grounding terminals **41**, a plurality of second power terminals and a plurality of second signal terminals. Each of the second grounding terminals **41**, each of the second power terminals and each of the second signal terminals are made of an electrical conducting material.

(19) The middle partition **50** is made of a metal material, for example copper, brass, nickel, silver, steel, tin or an alloy thereof, here is not intended to be limiting. The middle partition **50** is disposed in the tongue piece **12** and located between the first terminal row **30** and the second terminal row **40**. In some embodiments, the middle partition **50** is inserted in the flat slot **115** from the rear end **112** of the base **11**. As such, an electromagnetic shielding effect is provided between the first terminal row **30** and the second terminal row **40**, and signals between the first terminal row **30** and the second terminal **40** are prevented from being interfered.

(20) The first shielding sheet **60** is made of a metal material, for example copper, brass, nickel, silver, steel, tin or an alloy thereof, here is not intended to be limiting. The first shielding sheet **60** is inserted in the base **11** and disposed between the first terminal row **30** and the shielding case **20**. In some embodiments, the first shielding sheet **60** is inserted in the top part **113** of the base **11**, in other words the first shielding sheet **60** is located above the first terminal row **30**. In some embodiments, the first shielding sheet **60** may be inserted in the bottom part **114** of the base **11**. The first shielding sheet **60** has a plurality of first terminal elastic arms **61** and a first case elastic arm **62**. Each of the first terminal elastic arms **61** is inclined and extended from the first shielding sheet **60** toward the first terminal row **30** and elastically abuts against each of the first grounding terminals **31** to form an electric connection, in other words each of the first terminal elastic arms **61** is downwardly inclined and extended to the first terminal row **30** to elastically abut against each of the first grounding terminals **31**. The first case elastic arm **62** is inclined and extended from the first shielding sheet **60** toward the shielding case **20** and elastically abuts against the shielding case **20** to form an electric connection, in other words the first case elastic arm **62** is upwardly inclined and extended to elastically abut against an inner wall of the top part **113** of the shielding case **20**.

(21) The second shielding sheet **70** is made of a metal material, for example copper, brass, nickel, silver, steel, tin or an alloy thereof, here is not intended to be limiting. The second shielding sheet **70** is inserted in the base **11** and disposed between the second terminal row **40** and the shielding case **20**. In some embodiments, the second shielding sheet **70** is inserted in the bottom part **114** of the base **11**, in other words the second shielding sheet **70** is located below the second terminal row **40**. In some embodiments, the second shielding sheet **70** may be inserted in the top part **113** of the base **11**. The second shielding sheet **70** has a plurality of second terminal elastic arms **71** and a second case elastic arm **72**. Each of the second terminal elastic arms **71** is inclined and extended from the second shielding sheet **70** toward the second terminal row **40** and elastically abuts against each of the second grounding terminals **41** to form an electric connection, in other words each of the second terminal elastic arms **71** is upwardly inclined and extended to the second terminal row **40** to elastically abut against each of the second grounding terminals **41**. The second case elastic arm **72** is inclined and extended from the second shielding sheet **70** toward the shielding case **20** and elastically abuts against the shielding case **20** to form an electric connection, in other words the second case elastic arm **72** is downwardly inclined and extended to elastically abut against an inner wall of the bottom part **114** of the shielding case **20**.

(22) Accordingly, each of the first grounding terminals **31** of the first terminal row **30** abuts against each of the first terminal elastic arms **61** of the first shielding sheet **60**, and abuts against the shielding case **20** through the first case elastic arm **62** of the first shielding sheet **60**. Each of the second grounding terminals **41** of the second terminal row **40** abuts against each of the second terminal elastic arms **71** of the second shielding sheet **70**, and abuts against the shielding case **20** through the second case elastic arm **72** of the second shielding sheet **70**. As such, interferences of the electromagnetic wave and the crosstalk generated by the first terminal row **30** and the second terminal row **40** are transferred to the soldering feet **21** of the shielding case **20** and the circuit board, thus the stability during the high-frequency and high-speed transmission is increased.

(23) Details are provided as follows. Please refer to FIG. 3 and FIG. 5, the first shielding sheet **60** has a partition elastic arm **63**. The partition elastic arm **63** is inclined and extended from one side of the first shielding sheet **60** toward the middle partition **50** to elastically abut against the middle partition **50** to form an electric connection, in other words the partition elastic arm **63** is downwardly inclined and extended and abuts against the middle partition **50**. As such, the electromagnetic wave shielding generated by the middle partition **50** is transferred to the circuit board sequentially through the partition elastic arm **63**, the first shielding sheet **60**, the first case elastic arm **62**, the shielding case **20** and the soldering feet **21**, thus the stability during the high-frequency and high-speed transmission is effectively increased.

(24) Please refer to FIG. 2, FIG. 3, FIG. 6, and FIG. 8, the top part **113** of the base **11** has a first accommodating slot **116** and at least one first positioning slot **117**. The first shielding sheet **60** has at least one first positioning sheet **64**. The amount of the first positioning sheet **64** is corresponding to the amount of first positioning slot **117**. The first positioning sheet **64** is bent and extended out from the first shielding sheet **60**. The first shielding sheet **60** is accommodated in the first accommodating slot **116**. The first positioning sheet **64** is inserted and positioned in the first positioning slot **117**. In some embodiments, there are three first positioning slots **117** and three first positioning sheets **64**, here is not intended to be limiting. For example, the amount of the first positioning slot **117** and the amount of the first positioning sheet **64** are two, or equal to or more than four. In some embodiments, one of the first positioning slots **117** and the other two first positioning slots **117** are disposed at two opposite sides of the first accommodating slot **116**, one of the first positioning sheets **64** and the other two first positioning sheets **64** are disposed at two opposite sides of the first shielding sheet **60**. As such, each of the first positioning sheets **64** is inserted and positioned in each of the first positioning slots **117**, thus the first shielding sheet **60** is fastened in the first accommodating slot **116**.

(25) Please refer to FIG. 2, FIG. 4, FIG. 7, and FIG. 9, the bottom part **114** of the base **11** has a second accommodating slot **118** and at least one second positioning slot **119**. The second shielding sheet **70** has at least one second positioning sheet **73**. The amount of the second positioning sheet **73** is corresponding to the amount of second positioning slot **119**. The second positioning sheet **73** is bent and extended out from the second shielding sheet **70**. The second shielding sheet **70** is accommodated in the second accommodating slot **118**. The second positioning sheet **73** is inserted and positioned in the second positioning slot **119**. In some embodiments, there is one second positioning slot **119** and one second positioning sheets **73**, here is not intended to be limiting. For example, the amount of the second positioning slot **119** and the amount of the second positioning sheet **73** are equal to or more than two. In some embodiments, the second accommodating slot **118** and the second positioning slot **119** are in communication and adjacently disposed. The base **11** is extended with a block wall **110** arranged between the second accommodating slot **118** and the second positioning slot **119**. The block wall **110** blocks one lateral surface of the second positioning sheet **73**, thus the block wall **110** and the second positioning slot **119** jointly cover the a front side, a rear side, a left side and a right side of the second positioning sheet **73**. In some embodiments, there are two block walls **110** which are oppositely arranged as shown in FIG. 4 and FIG. 7, thus an optimal blocking effect is provided to the second positioning sheet **73**.

(26) Please refer to FIG. 4, each of the second terminal elastic arms **71** and the second positioning sheet **73** are disposed at two opposite sides of the second shielding sheet **70**. As such, the second positioning sheet **73** is restrained between the block wall **110** and the second positioning slot **119**, and the second shielding sheet **70** generates an upward pivotal torque, the second shielding sheet **70** is tightly attached in the second accommodating slot **118**, and each of the second terminal elastic arms **71** tightly and correspondingly abuts against each of the second grounding terminals **41**, thus the abutting connection between each of the second terminal elastic arms **71** and each of the second grounding terminals **41** is greatly ensured as shown in FIG. 8 and FIG. 9.

(27) Please refer to FIG. 2, FIG. 3, FIG. 4, FIG. 6, and FIG. 7, in some embodiments, there are two first case elastic arms **62** and two second case elastic arms **72**, here is not intended to be limiting. Only one of the first case elastic arms **62** and only one of the second case elastic arms **72** may be adopted as long as the transferring effect is achieved. In some embodiments, there are two first case elastic arms **62** and two second case elastic arms **72** adopted to achieve a desirable transferring effect in a limited space, thus the amount of the first case elastic arm **62** and the amount of the second case elastic arm **72** may be equal to or more than two. Each of the first case elastic arms **62** and each of the first terminal elastic arms **61** are alternately arranged. Each of the second case elastic arms **72** and each of the second terminal elastic arms **71** are alternately arranged.

(28) According to the electrical connector structure of the present disclosure, the electric connection is formed through each of the first terminal elastic arms **61** and the first case elastic arm **62** of the first shielding sheet **60** respectively abutting against each of the first grounding terminals **31** and the shielding case **20**, and the electric connection is formed through each of the second terminal elastic arms **71** and the second case elastic arm **72** of the second shielding sheet **70** respectively abutting against each of the second grounding terminals **41** and the shielding case **20**, the interference of the electromagnetic wave and the remote crosstalk is still effectively improved under a situation of the first terminal row **30**, the second terminal row **40** and the middle partition **50** are distanced from the shielding case **20**, thus the stability during the high-frequency and high-speed transmission is increased. Moreover, the first terminal row **30**, the first shielding sheet **60**, the middle partition **50**, the second terminal row **40** and the second shielding sheet **70** are inserted and assembled from the rear end **112** of the base **11**, thus the manufacturing procedure does not require the injection molding and mounting process, the assembling production is simplified and the production cost is reduced.

(29) While this disclosure has been described by means of specific embodiments, numerous modifications and variations could be made thereto by those skilled in the art without departing from the scope and spirit of this disclosure set forth in the claims.

Claims

1. An electrical connector structure, comprising: an insulation body, comprising a base and a tongue piece, wherein the base comprises a front end and a rear end, the tongue piece is extended out from the front end, the tongue piece comprises a first distal surface and a second distal surface oppositely arranged; a shielding case, adapted to sheathe the insulation body; a first terminal row, inserted in the insulation body from the rear end and disposed on the first distal surface, and comprising a plurality of first grounding terminals; a second terminal row, inserted in the insulation body from the rear end and disposed on the second distal surface, and comprising a plurality of second grounding terminals; a middle partition, accommodated in the tongue piece and located between the first terminal row and the second terminal row; a first shielding sheet, inserted in the base and located between the first terminal row and the shielding case, comprising a plurality of first terminal elastic arms and a first case elastic arm, wherein each of the first terminal elastic arms is extended from the first shielding sheet toward the first terminal row and abuts against each of the first grounding terminals, and the first case elastic arm is extended from the first shielding sheet

toward the shielding case and abuts against the shielding case; and a second shielding sheet, inserted in the base and located between the second terminal row and the shielding case, comprising a plurality of second terminal elastic arms and a second case elastic arm, wherein each of the second terminal elastic arms is extended from the second shielding sheet toward the second terminal row and abuts against each of the second grounding terminals, and the second case elastic arm is extended from the second shielding sheet toward the shielding case and abuts against the shielding case; wherein the base comprises a flat slot inwardly extended from the rear end and disposed in the tongue piece, and the middle partition is inserted in the flat slot.

2. The electrical connector structure according to claim 1, wherein the first shielding sheet comprises a partition elastic arm, and the partition elastic arm is extended from the first shielding sheet toward the middle partition and abuts against the middle partition.

3. The electrical connector structure according to claim 1, wherein the first case elastic arm is multiple in number, and each of the first case elastic arms and each of the first terminal elastic arms are alternately arranged.

4. The electrical connector structure according to claim 1, wherein the second case elastic arm is multiple in number, and each of the second case elastic arms and each of the second terminal elastic arms are alternately arranged.

5. The electrical connector structure according to claim 1, wherein the base comprises a first accommodating slot and a first positioning slot, the first shielding sheet comprises a first positioning sheet, the first shielding sheet is accommodated in the first accommodating slot, and the first positioning sheet is inserted and positioned in the first positioning slot.

6. The electrical connector structure according to claim 5, wherein a number of the first positioning slot is three and a number of the first positioning sheet is three, one of the first positioning slots and other two first positioning slots are respectively disposed at two sides of the first accommodating slot opposite to each other, one of the first positioning sheets and other two first positioning sheets are respectively disposed at two sides of the first shielding sheet opposite to each other, and each of the first positioning sheets is inserted and positioned in each of the first positioning slots.

7. The electrical connector structure according to claim 1, wherein the base comprises a second accommodating slot and a second positioning slot, the second shielding sheet comprises a second positioning sheet, the second shielding sheet is accommodated in the second accommodating slot, and the second positioning sheet is inserted and positioned in the second positioning slot.

8. The electrical connector structure according to claim 7, wherein the second accommodating slot and the second positioning slot communicates with each other and are adjacently disposed, the base comprises a block wall arranged between the second accommodating slot and the second positioning slot, and the block wall blocks a lateral surface of the second positioning sheet.

9. The electrical connector structure according to claim 7, wherein each of the second terminal elastic arms and the second positioning sheet are disposed at two sides of the second shielding sheet.
